

OLDENBURG, Germany

WMO: 102150

Lat: 53.183N

Lon: 8.167E

Elev: 20

StdP: 101.08

Time Zone: 1.00 (EUC)

Period: 94-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.7	-11.8	1.4	-7.3	-9.5	1.7	-5.6	12.2	6.3	10.6	5.4	3.3	90	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	29.4	19.3	27.1	18.7	24.9	17.6	20.5	26.7	19.5	25.1	18.6	23.6	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.3	22.6	17.4	12.5	21.5	16.6	11.8	20.7	58.7	26.8	55.4	25.2	52.6	23.6	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.2	7.2	-11.7	33.0	2.8	1.7	-13.7	34.2	-15.3	35.2	-16.9	36.2	-18.9	37.5		
(5)			-12.0	22.5	2.8	0.9	-14.0	23.2	-15.6	23.7	-17.2	24.2	-19.2	24.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	2.2	2.7	5.2	9.3	12.9	15.6	18.3	18.1	14.5	10.2	6.0	2.2		
	DBStd	6.90	4.52	4.31	3.60	3.86	3.45	3.31	3.28	3.04	2.91	3.53	3.61	4.56		
	HDD10.0	1075	241	205	151	58	12	0	0	0	1	41	125	241		
	HDD18.3	3205	499	438	406	272	172	95	43	42	120	251	369	499		
	CDD10.0	1012	1	1	4	37	102	169	256	251	137	48	6	1		
	CDD18.3	98	0	0	0	1	3	14	41	34	6	0	0	0		
	CDH23.3	831	0	0	0	8	41	132	342	268	40	1	0	0		
	CDH26.7	233	0	0	0	0	6	34	107	82	4	0	0	0		
(14) Wind	WSAvg	3.3	3.8	3.9	4.0	3.4	3.2	3.1	3.0	2.9	2.9	3.1	3.2	3.4		
(15) Precipitation	PrecAvg	813	70	58	55	44	61	74	87	77	74	64	71	78		
	PrecMax	1126	139	117	123	99	110	154	180	175	192	193	146	179		
	PrecMin	625	6	16	9	2	20	24	29	24	14	13	2	19		
	PrecStd	119	37	27	32	23	25	35	37	34	41	34	36	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.3	13.2	17.3	24.6	27.3	30.2	32.5	32.0	27.0	21.0	15.5	12.8		
		MCWB	10.2	9.8	10.8	14.0	17.7	19.6	20.0	19.8	18.6	16.5	12.4	11.3		
	2%	DB	10.8	11.4	14.5	21.2	24.1	27.0	29.6	29.1	24.1	18.3	13.7	11.1		
		MCWB	9.1	9.1	10.0	13.1	15.5	18.3	19.9	18.9	18.0	14.8	12.0	9.8		
	5%	DB	9.2	9.9	12.4	18.6	21.8	24.3	27.2	26.3	21.4	16.6	12.3	9.7		
		MCWB	7.8	8.3	9.2	11.9	14.7	17.2	19.1	18.8	16.3	13.9	10.6	8.3		
	10%	DB	8.0	8.5	10.8	16.1	19.6	21.8	24.7	23.9	19.3	15.2	11.0	8.4		
		MCWB	6.7	7.0	8.1	10.9	13.5	16.3	18.0	17.8	15.2	12.9	9.4	7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.8	10.9	12.3	15.4	18.8	20.6	22.2	21.7	19.7	16.6	13.3	11.4		
		MCDB	11.8	12.5	15.6	21.0	25.2	27.7	29.0	28.3	25.1	20.2	14.4	12.4		
	2%	WB	9.4	9.5	10.8	13.8	16.8	19.3	20.8	20.1	18.0	15.3	12.1	9.9		
		MCDB	10.6	10.8	13.1	19.3	22.5	25.2	27.2	25.9	23.2	17.6	13.6	11.1		
	5%	WB	8.0	8.3	9.7	12.7	15.3	18.2	19.6	19.1	16.9	14.2	10.7	8.5		
		MCDB	9.0	9.7	12.1	17.4	19.8	22.7	25.4	24.2	20.7	16.0	12.0	9.5		
	10%	WB	6.8	7.2	8.3	11.4	14.1	16.9	18.6	18.3	15.8	13.3	9.6	7.2		
		MCDB	7.9	8.5	10.5	15.7	18.5	20.9	23.7	23.0	18.5	14.9	10.7	8.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.2	7.0	9.2	9.4	9.0	8.9	8.9	8.1	7.0	4.9	4.3		
		MCDBR	4.9	5.7	9.8	13.5	13.4	12.9	13.4	13.0	11.1	8.1	5.5	4.9		
	5% WB	MCWBR	4.3	4.6	6.3	6.8	6.7	6.1	5.6	5.4	5.7	5.2	4.2	4.5		
		MCWBR	4.9	5.1	8.6	11.4	11.4	11.6	12.2	10.8	10.0	7.4	5.2	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.345	0.376	0.409	0.406	0.404	0.419	0.412	0.387	0.360	0.339	0.318		
		taud	2.323	2.326	2.259	2.219	2.262	2.290	2.262	2.296	2.356	2.398	2.352	2.314		
	Ebn at Noon		641	749	800	820	843	847	825	809	783	723	617	573		
		Edh at Noon	61	82	106	125	126	124	126	115	96	75	59	52		
(44) All-Sky Solar Radiation	RadAvg		0.59	1.27	2.43	4.02	4.98	5.24	5.00	4.16	3.00	1.69	0.74	0.46		
	RadStd		0.06	0.18	0.27	0.46	0.50	0.40	0.64	0.32	0.34	0.17	0.07	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-128	N/A	N/A	

Nomenclature: See separate page